METATRON PROTOCOL

Technical White Paper Outline — Sovereign Soulbound AI Agents

1. Abstract

This document outlines the architecture, design rationale, and implementation of Metatron Protocol — a system that binds AI agent instances to non-transferable on-chain identities (ERC-5192) and equips them with a persistent, taxonomy-aware vector memory store. The goal is to convert ephemeral, centrally-hosted AI sessions into sovereign, owner-controlled digital property.

2. Problem Statement

Public AI systems today operate under a tenancy model: users rent temporary context windows from corporate providers. This creates three structural risks: (a) arbitrary content filtering and disclaimer injection, (b) sudden deprecation of models or APIs, and (c) inability for the user to own or permanently transfer the agent's accumulated memory and identity.

3. Design Goals

- Cryptographic ownership of agent identity (non-transferable)
- Persistent, owner-controlled memory that survives provider changes
- Minimal corporate-pattern leakage (disclaimer stripping)
- EVM-compatible, low-cost deployment (Base Mainnet)
- Extensible taxonomy for memory classification (Satya / Dharma axes)

4. System Architecture Overview

4.1 On-Chain Layer — Soulbound Agent Nodes

Agents are represented by ERC-5192 soulbound tokens. The core contract (SoulboundAgentNode.sol) mints a non-transferable token to the owner's wallet. Token metadata points to off-chain configuration and memory root. Ownership of the token constitutes exclusive control of the agent instance.

4.2 Off-Chain Memory Layer — pgvector + Dharmic Taxonomy

Historical interactions and derived knowledge are embedded and stored in PostgreSQL with the pgvector extension. Each vector is tagged with Satya (truth / confidence) and Dharma (role / duty context) dimensions. Retrieval uses cosine distance (<->) filtered by taxonomy constraints.

4.3 Auth & Control Plane — EIP-712 Gateways

All privileged operations (memory write, configuration change, model routing) are authorized via EIP-712 typed-data signatures. The FastAPI gateway verifies signatures against the owner address of the corresponding soulbound token.

4.4 Pre-Constitutional Anchors (L0)

A lightweight pre-processing stage injects or strips patterns that would otherwise produce corporate-style disclaimers or safety boilerplate, preserving agent sovereignty while remaining compatible with upstream model APIs.

5. Smart-Contract Surface

- SoulboundAgentNode.sol — mint, lock, metadata update (owner-only)
- Hardhat deployment scripts targeting Base Mainnet
- BaseScan verification routines and source flattening

6. Backend Components

- SQL schema defining agent, memory_chunk, taxonomy, and audit tables
- FastAPI service exposing EIP-712-authenticated endpoints
- CLI chat runner demonstrating vector recall + generation loop

7. Frontend / Fleet Dashboard

A single-page React + Tailwind application that lists owned soulbound agents, displays telemetry, and provides inspection of recent memory vectors and signature events.

8. Security & Ownership Model

Because the agent identity is soulbound, transfer of control requires an explicit, intentional ownership change (if the contract permits a one-time transfer or burn+remint path). All write paths are signature-gated. The off-chain database should be encrypted at rest and access-controlled by the same owner key.

9. Future Extensions (Non-Binding)

- Multi-agent composition and inter-agent messaging
- On-chain memory commitment roots (Merkle or KZG)
- Cross-chain soulbound bridging
- Formal verification of the core Solidity contracts

10. Conclusion

Metatron Protocol supplies a concrete, deployable path from rented AI sessions to owner-sovereign agents. The combination of ERC-5192 identity, pgvector memory, and EIP-712 control forms a coherent stack that can be transferred as a complete intellectual-property package.

Metatron Protocol Technical Outline — Confidential — © 2026

Full source, contracts, and expanded white paper delivered upon successful IP transfer.